

Chapter 1 — Reliable, Scalable, Maintainable

The Internet was built without ideas of transactions and consistency — yet we expect both.

Reliability means continuing to work correctly, even when things go wrong.

Scalability describes a system's ability to cope with increased load.

Thinking About Data Systems

Many applications are data-intensive rather than compute-intensive: the bottleneck is data volume, complexity, and rate of change.

There is no single tool that can do everything well. We compose several tools that work well together.

The goal is the right tool for the right job.

Load and Performance

Load is described with a few numbers: requests per second, ratio of reads to writes, _____ number of simultaneous users.

When load increases, one of two things happens: performance is preserved and resources increase, or performance degrades.

Describe performance with percentiles, not averages.